BRAC UNIVERSITY

Inspiring Excellence

Department of Mathematics and Natural Sciences
Mid-Term Examination
Semester: Summer 2025
Course Title: MATHEMATICS I: Differential Calculus and Coordinate Geometry
Course Code: MAT- 110

Time: 1 hour 20 minutes **Date:** July 27, 2025
Instructions: **Total marks:** 40
[$5 \times 8 = 40$]

- Answer any **five** questions.
- The figures in the right margin indicate full marks.
- The total marks will be converted to 20.
- All your works must be inside the answer script. Otherwise, it will not be counted.

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1. Evaluate limit: [8]

$$\lim_{x \to 0} (1 + \tan x)^{\frac{1}{\sin x}}$$

2. Check the continuity and differentiability of the following function f(x) at $x = 0$: [8]

$$f(x) = \begin{cases} x^2 \cos\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$$

[Hints: Use the Squeeze/Sandwich Theorem if necessary]

3. Find the differential coefficientof the following function with respect to x: [8]

$$(\sin x)^{\cos x} + (\cos x)^{\sin x}$$

4. Find all critical and stationary points of the function $f(x) = 3x^{5/3} - 15x^{2/3}$. [8]
Also find the relative extrema.

5. Verify the hypothesis of the Mean Value Theorem for the function [8]

$$f(x) = \sqrt{4x - 3}$$

over the interval [1,3]. Also find the value of ‘c' in that interval that satisfy the conclusion of the theorem.

6. Find the Maclaurin polynomials P_0, P_1, P_2, P_3 of the function $f(x) = e^x \cos x$. [8]

7. Let, $f(x) = \frac{x-1}{\sqrt{x^2}}$ and $\lim_{x \to +\infty} f(x) = a$ and $\lim_{x \to -\infty} f(x) = b$. [8]
The straight lines $y = a$ and $y = b$ are called **HORIZONTAL ASYMPTOTES** of the graph of $y = f(x) = \frac{x-1}{\sqrt{x^2}}$.
Find the value of $(a - b)$.
Hints: $\sqrt{x^2} = |x| = \begin{cases} -x, & x < 0 \\ x, & x \geq 0 \end{cases}$

SOLUTION & MARKING SCHEME
(Created by: SAKAL ROY [SKY])

TO APPRECIATE YOUR MATHEMATICAL THINKING, ANY VALID MATHEMATICAL WAY OF SOLVING WILL BE TREATED EQUALLY AND ASSESSED ACCORDINGLY, AS LONG AS IT'S CONSISTENT WITH THE PROVIDED INSTRUCTIONS OF THE QUESTION.

IN THIS ASSESSMENT PROCESS EVERY PROBLEM WILL BE ASSESSED BASED ON 3 CRITERIA:
A. MATHEMATICAL THINKING
B. MATHEMATICAL KNOWLEDGE
C. MATHEMATICAL SKILL

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1. Evaluate limit: [8]

$$\lim_{x\to 0}(1+\tan x)^{\frac{1}{\sin x}}$$

Solution
See full soluion here: **https://www.emathhelp.net/calculators/calculus-1/limit-calculator/?f=%281%2B+tan+%28x%29%29%5E%281%2Fsin%28x%29%29&var=&val=0&dir=**
Since $u = e^{\ln(u)}$, then: $$\lim_{x\to 0}(\tan(x)+1)^{\frac{1}{\sin(x)}} = \lim_{x\to 0} e^{\ln\left((\tan(x)+1)^{\frac{1}{\sin(x)}}\right)}$$ Simplify: $$\lim_{x\to 0} e^{\ln\left((\tan(x)+1)^{\frac{1}{\sin(x)}}\right)} = \lim_{x\to 0} e^{\frac{\ln(\tan(x)+1)}{\sin(x)}}$$ Move the limit under the exponential: $$\lim_{x\to 0} e^{\frac{\ln(\tan(x)+1)}{\sin(x)}} = e^{\lim_{x\to 0}\frac{\ln(\tan(x)+1)}{\sin(x)}}$$

Since we have an indeterminate form of type $\frac{0}{0}$, we can apply the l'Hopital's rule:

$$e^{\lim_{x\to 0} \frac{\ln(\tan(x)+1)}{\sin(x)}} = e^{\lim_{x\to 0} \frac{\frac{d}{dx}(\ln(\tan(x)+1))}{\frac{d}{dx}(\sin(x))}}$$

For steps, see derivative calculator.

$$e^{\lim_{x\to 0} \frac{\frac{d}{dx}(\ln(\tan(x)+1))}{\frac{d}{dx}(\sin(x))}} = e^{\lim_{x\to 0} \frac{\tan^2(x)+1}{(\tan(x)+1)\cos(x)}}$$

Substitute the variable with the value:

$$e^{\lim_{x\to 0} \frac{\tan^2(x)+1}{(\tan(x)+1)\cos(x)}} = e^{1}$$

Therefore,

$$\lim_{x\to 0} (\tan(x)+1)^{\frac{1}{\sin(x)}} = e$$

Answer: $\lim_{x\to 0} (\tan(x)+1)^{\frac{1}{\sin(x)}} = e$

2. Check the continuity and differentiability of the following function f(x) at x = 0: [8]

$$f(x) = \begin{cases} x^2\cos\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$$

[Hints: Use the Squeeze/Sandwich Theorem if necessary]

Solution
Continuity
To test the **continuity** of $\mathbf{f(x)}$ **at** $\mathbf{x = a}$, We've to explore the following: $$\lim_{x\to a} f(x) = f(a)$$ Here, $\mathbf{a = 0}$. Check the continuity of f(x) at x =0 by verifying $\lim_{x\to 0} f(x) = f(0)$. You can do that by finding the values of $\lim_{x\to 0^+} f(x)$, $\lim_{x\to 0^-} f(x)$ and $f(0)$ separately. We've to find: $\lim_{x\to 0} f(x)$ First we've to find: Right Hand Limit $(R.H.L) = \lim_{x\to 0^+} f(x)$
Method 1: $\lim_{x\to 0^+} f(x) = \lim_{x\to 0^+}\left[x^2\cos\left(\frac{1}{x}\right)\right]$ Now, $\lim_{x\to 0^+} x^2 = 0$. But, $\lim_{x\to 0^+}\left[\cos\left(\frac{1}{x}\right)\right]$ doesn't exist. So, we can't use limit laws for product of two limits here. Therefore, we'll use Squeeze/Sandwich theorem to determine $\lim_{x\to 0^+}\left[x^2\cos\left(\frac{1}{x}\right)\right]$. Now, we know: for $x \neq 0$: $-1 \leq \cos\left(\frac{1}{x}\right) \leq 1$ $\Rightarrow -x^2 \leq x^2\cos\left(\frac{1}{x}\right) \leq x^2$ [since, for, $x \to 0^+, x^2 > 0$] $\Rightarrow \lim_{x\to 0^+}(-x^2) \leq \lim_{x\to 0^+}\left[x^2\cos\left(\frac{1}{x}\right)\right] \leq \lim_{x\to 0^+}(x^2)$ $\Rightarrow -0 \leq \lim_{x\to 0^+}\left[x^2\cos\left(\frac{1}{x}\right)\right] \leq 0$ $\Rightarrow 0 \leq \lim_{x\to 0^+}\left[x^2\cos\left(\frac{1}{x}\right)\right] \leq 0$ Now, according to Squeeze/Sandwich theorem: $\mathbf{\lim_{x\to 0^+}\left[x^2\cos\left(\frac{1}{x}\right)\right] = 0}$ **So,** $\mathbf{\lim_{x\to 0^+} f(x) = 0}$
Method 2: $\lim_{x\to 0^+} f(x)$ Let, $x = 0 + h$, where $h \to 0^+$ So, $\lim_{x\to 0^+} f(x) = \lim_{h\to 0^+} f(0+h)$ Now, $0 + h > 0$ since $h \to 0^+$ So, $\lim_{h\to 0^+} f(0+h) = \lim_{h\to 0^+}\left[(0+h)^2\cos\left(\frac{1}{0+h}\right)\right] = \lim_{h\to 0^+}\left[h^2\cos\left(\frac{1}{h}\right)\right]$ Now, $\lim_{h\to 0^+} h^2 = 0$. But, $\lim_{h\to 0^+}\left[\cos\left(\frac{1}{h}\right)\right]$ doesn't exist. So, we can't use limit laws for product of two limits here. Therefore, we'll use Squeeze/Sandwich theorem to determine $\lim_{h\to 0^+}\left[h^2\cos\left(\frac{1}{h}\right)\right]$. Now, we know: for $h \neq 0$: $-1 \leq \cos\left(\frac{1}{h}\right) \leq 1$ $\Rightarrow -h^2 \leq h^2\cos\left(\frac{1}{h}\right) \leq h^2$ [since, for $h \to 0^+, h^2 > 0$]

$\Rightarrow \lim\limits_{h\to0^+}(-h^2) \le \lim\limits_{h\to0^+}\left[h^2\cos\left(\frac{1}{h}\right)\right] \le \lim\limits_{h\to0^+}(h^2)$

$\Rightarrow -0 \le \lim\limits_{h\to0^+}\left[h^2\cos\left(\frac{1}{h}\right)\right] \le 0$

$\Rightarrow 0 \le \lim\limits_{h\to0^+}\left[h^2\cos\left(\frac{1}{h}\right)\right] \le 0$

Now, according to Squeeze/Sandwich theorem: $\mathbf{\lim\limits_{h\to0^+}\left[h^2\cos\left(\frac{1}{h}\right)\right] = 0}$

So, $\mathbf{\lim\limits_{x\to0^+} f(x) = 0}$

Now, we've to find: Left Hand Limit (L. H. L) $= \lim\limits_{x\to0^-} f(x)$

Method 1:

$\lim\limits_{x\to0^-} f(x) = \lim\limits_{x\to0^-}\left[x^2\cos\left(\frac{1}{x}\right)\right]$

Now, $\lim\limits_{x\to0^-} x^2 = 0$. But, $\lim\limits_{x\to0^-}\left[\cos\left(\frac{1}{x}\right)\right]$ doesn't exist.

So, we can't use limit laws for product of two limits here.

Therefore, we'll use Squeeze/Sandwich theorem to determine $\lim\limits_{x\to0^-}\left[x^2\cos\left(\frac{1}{x}\right)\right]$.

Now, we know: for $x \ne 0$:

$-1 \le \cos\left(\frac{1}{x}\right) \le 1$

$\Rightarrow -x^2 \le x^2\cos\left(\frac{1}{x}\right) \le x^2$ [since, for, $x \to 0^-, x^2 > 0$]

$\Rightarrow \lim\limits_{x\to0^-}(-x^2) \le \lim\limits_{x\to0^-}\left[x^2\cos\left(\frac{1}{x}\right)\right] \le \lim\limits_{x\to0^-}(x^2)$

$\Rightarrow -0 \le \lim\limits_{x\to0^-}\left[x^2\cos\left(\frac{1}{x}\right)\right] \le 0$

$\Rightarrow 0 \le \lim\limits_{x\to0^-}\left[x^2\cos\left(\frac{1}{x}\right)\right] \le 0$

Now, according to Squeeze/Sandwich theorem: $\mathbf{\lim\limits_{x\to0^-}\left[x^2\cos\left(\frac{1}{x}\right)\right] = 0}$

So, $\mathbf{\lim\limits_{x\to0^-} f(x) = 0}$

Method 2:

$\lim\limits_{x\to0^-} f(x)$

Let, $x = 0 - h$, where $h \to 0^+$

So, $\lim\limits_{x\to0^-} f(x) = \lim\limits_{h\to0^+} f(0-h)$

Now, $0 - h < 0$ since $h \to 0^+$

So, $\lim\limits_{h\to0^+} f(0-h) = \lim\limits_{h\to0^+}\left[(0-h)^2\cos\left(\frac{1}{0-h}\right)\right] = \lim\limits_{h\to0^+}\left[(-h)^2\cos\left(\frac{1}{-h}\right)\right] = \lim\limits_{h\to0^+}\left[h^2\cos\left(\frac{1}{h}\right)\right]$

Now, $\lim\limits_{h\to0^+} h^2 = 0$. But, $\lim\limits_{h\to0^+}\left[\cos\left(\frac{1}{h}\right)\right]$ doesn't exist.

So, we can't use limit laws for product of two limits here.

Therefore, we'll use Squeeze/Sandwich theorem to determine $\lim\limits_{h\to0^+}\left[h^2\cos\left(\frac{1}{h}\right)\right]$.

Now, we know: for $h \ne 0$:

$-1 \le \cos\left(\frac{1}{h}\right) \le 1$

$\Rightarrow -h^2 \le h^2\cos\left(\frac{1}{h}\right) \le h^2$ [since, for $h \to 0^+, h^2 > 0$]

$\Rightarrow \lim\limits_{h\to0^+}(-h^2) \le \lim\limits_{h\to0^+}\left[h^2\cos\left(\frac{1}{h}\right)\right] \le \lim\limits_{h\to0^+}(h^2)$

$\Rightarrow -0 \le \lim\limits_{h\to0^+}\left[h^2\cos\left(\frac{1}{h}\right)\right] \le 0$

$\Rightarrow 0 \le \lim\limits_{h\to0^+}\left[h^2\cos\left(\frac{1}{h}\right)\right] \le 0$

Now, according to Squeeze/Sandwich theorem: $\mathbf{\lim\limits_{h\to0^+}\left[h^2\cos\left(\frac{1}{h}\right)\right] = 0}$

So, $\mathbf{\lim\limits_{x\to0^-} f(x) = 0}$

Now, $\mathbf{f(0) = 0}$.

If $\mathbf{\lim\limits_{x\to a^+} f(x) = \lim\limits_{x\to a^-} f(x) = f(a)}$ **then** $\mathbf{f(x)}$ **is continuous at** $\mathbf{x = a}$**.**

And if all of $\lim_{x\to a^+} f(x)$, $\lim_{x\to a^-} f(x)$ and $f(a)$ aren't equal then $f(x)$ is not continuous at $x = a$

Here, $a = 0$, then: $\lim_{x\to 0^+} f(x) = \lim_{x\to 0^-} f(x) = f(0)$.

So, $f(x)$ is continuous at $x = 0$.

Differentiability

To test the **differentiability** of $\mathbf{f(x)}$ **at** $\mathbf{x = a}$, We've to explore the following:

$$\mathbf{f'(a) = \lim_{x\to a} \frac{f(x) - f(a)}{x - a} = \lim_{x\to a} \frac{f(a) - f(x)}{a - x} = \lim_{h\to 0} \frac{f(a + h) - f(a)}{h} = \lim_{h\to 0} \frac{f(a) - f(a - h)}{h}}$$

Here, let, $f(x) = \begin{cases} x^2\cos\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$ and $a = 0$

So, We've to verify:

$$f'(0) = \lim_{x\to 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x\to a} \frac{f(0) - f(x)}{0 - x} = \lim_{h\to 0} \frac{f(0 + h) - f(0)}{h} = \lim_{h\to 0} \frac{f(0) - f(0 - h)}{h}$$

Since, $f(x) = \begin{cases} x^2\cos\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$

So, $\mathbf{f(0) = 0}$

$$\Rightarrow \mathbf{f'(0) = \lim_{x\to 0} \frac{f(x) - 0}{x - 0} = \lim_{x\to 0} \frac{0 - f(x)}{0 - x} = \lim_{h\to 0} \frac{f(0 + h) - 0}{h} = \lim_{h\to 0} \frac{0 - f(0 - h)}{h}}$$

First we've to find: Right Hand Derivative (R. H. D)

$$\mathbf{f'_+(0) = \lim_{x\to 0^+} \frac{f(x) - 0}{x - 0} = \lim_{x\to 0^+} \frac{0 - f(x)}{0 - x} = \lim_{h\to 0^+} \frac{f(0 + h) - 0}{h} = \lim_{h\to 0^-} \frac{0 - f(0 - h)}{h}}$$

Method 1:

$\mathbf{f'_+(0) = \lim_{x\to 0^+} \frac{f(x)-0}{x-0} = \lim_{x\to 0^+} \frac{0-f(x)}{0-x} = \lim_{x\to 0^+} \frac{f(x)}{x}}$

Now, since, $f(x) = x^2\cos\left(\frac{1}{x}\right)$, when $x \neq 0$.

So, for $x \to 0^+$, $f(x) = x^2\cos\left(\frac{1}{x}\right)$

So, $\lim_{x\to 0^+} \frac{f(x)}{x} = \lim_{x\to 0^+} \frac{x^2\cos\left(\frac{1}{x}\right)}{x} = \lim_{x\to 0^+} \left[x\cos\left(\frac{1}{x}\right)\right]$

Now, $\lim_{x\to 0^+} x = 0$. But, $\lim_{x\to 0^+} \left[\cos\left(\frac{1}{x}\right)\right]$ doesn't exist.

So, we can't use limit laws for product of two limits here.

Therefore, we'll use Squeeze/Sandwich theorem to determine $\lim_{x\to 0^+} \left[x\cos\left(\frac{1}{x}\right)\right]$.

Now, we know: for $x \neq 0$:

$-1 \leq \cos\left(\frac{1}{x}\right) \leq 1$

$\Rightarrow -x \leq x\cos\left(\frac{1}{x}\right) \leq x$ [since, for, $x \to 0^+, x > 0$]

$\Rightarrow \lim_{x\to 0^+} (-x) \leq \lim_{x\to 0^+} \left[x\cos\left(\frac{1}{x}\right)\right] \leq \lim_{x\to 0^+} (x)$

$\Rightarrow -0 \leq \lim_{x\to 0^+} \left[x\cos\left(\frac{1}{x}\right)\right] \leq 0$

$\Rightarrow 0 \leq \lim_{x\to 0^+} \left[x\cos\left(\frac{1}{x}\right)\right] \leq 0$

Now, according to Squeeze/Sandwich theorem: $\mathbf{\lim_{x\to 0^+} \left[x\cos\left(\frac{1}{x}\right)\right] = 0}$

So, $\mathbf{f'_+(0) = \lim_{x\to 0^+} \frac{f(x)-0}{x-0} = \lim_{x\to 0^+} \frac{0-f(x)}{0-x} = \lim_{x\to 0^+} \frac{f(x)}{x} = 0}$

Method 2:

$\mathbf{f'_+(0) = \lim_{h\to 0^+} \frac{f(0+h)-0}{h} = \lim_{h\to 0^+} \frac{f(h)}{h}}$

Now, since, $f(x) = x^2\cos\left(\frac{1}{x}\right)$, when $x \neq 0$.

So, for $h \to 0^+, f(h) = h^2 \cos\left(\frac{1}{h}\right)$

So, $\lim\limits_{h\to 0^+} \frac{f(h)}{h} = \lim\limits_{h\to 0^+} \frac{h^2\cos\left(\frac{1}{h}\right)}{h} = \lim\limits_{h\to 0^+} \left[h\cos\left(\frac{1}{h}\right)\right]$

Now, $\lim\limits_{h\to 0^+} h = 0$. But, $\lim\limits_{h\to 0^+} \left[\cos\left(\frac{1}{h}\right)\right]$ doesn't exist.

So, we can't use limit laws for product of two limits here.

Therefore, we'll use Squeeze/Sandwich theorem to determine $\lim\limits_{h\to 0^+} \left[h\cos\left(\frac{1}{h}\right)\right]$.

Now, we know: for $h \neq 0$:

$-1 \leq \cos\left(\frac{1}{h}\right) \leq 1$

$\Rightarrow -h \leq h\cos\left(\frac{1}{h}\right) \leq h$ [since, for $h \to 0^+, h > 0$]

$\Rightarrow \lim\limits_{h\to 0^+} (-h) \leq \lim\limits_{h\to 0^+} \left[h\cos\left(\frac{1}{h}\right)\right] \leq \lim\limits_{h\to 0^+} (h)$

$\Rightarrow -0 \leq \lim\limits_{h\to 0^+} \left[h\cos\left(\frac{1}{h}\right)\right] \leq 0$

$\Rightarrow 0 \leq \lim\limits_{h\to 0^+} \left[h\cos\left(\frac{1}{h}\right)\right] \leq 0$

Now, according to Squeeze/Sandwich theorem: $\mathbf{\lim\limits_{h\to 0^+} \left[h\cos\left(\frac{1}{h}\right)\right] = 0}$

So, $\mathbf{f'_+(0) = \lim\limits_{h\to 0^+} \frac{f(0+h)-0}{h} = \lim\limits_{h\to 0^+} \frac{f(h)}{h} = 0}$

Method 3:

$\mathbf{f'_+(0) = \lim\limits_{h\to 0^-} \frac{0-f(0-h)}{h} = \lim\limits_{h\to 0^-} \frac{-f(-h)}{h}}$

Now, since, $f(x) = x^2\cos\left(\frac{1}{x}\right)$, when $x \neq 0$.

So, for $h \to 0^- \Rightarrow -h \to 0^+, f(-h) = (-h)^2\cos\left(\frac{1}{-h}\right)$

So, $\lim\limits_{h\to 0^-} \frac{-f(-h)}{h}$

$= \lim\limits_{h\to 0^-} \frac{-(-h)^2\cos\left(\frac{1}{-h}\right)}{h}$

$= \lim\limits_{h\to 0^-} \frac{-h^2\cos\left(\frac{1}{h}\right)}{h}$ [Since, $\cos(-\theta) = \cos\theta$]

$= \lim\limits_{h\to 0^-} \left[-h\cos\left(\frac{1}{h}\right)\right]$

Now, $\lim\limits_{h\to 0^-} (-h) = 0$. But, $\lim\limits_{h\to 0^-} \left[\cos\left(\frac{1}{h}\right)\right]$ doesn't exist.

So, we can't use limit laws for product of two limits here.

Therefore, we'll use Squeeze/Sandwich theorem to determine $\lim\limits_{h\to 0^-} \left[-h\cos\left(\frac{1}{h}\right)\right]$.

Now, we know: for $h \neq 0$:

$-1 \leq \cos\left(\frac{1}{h}\right) \leq 1$

$\Rightarrow -(-h) \leq -h\cos\left(\frac{1}{h}\right) \leq -h$ [since, for $h \to 0^- \Rightarrow -h \to 0^+, -h > 0$]

$\Rightarrow \lim\limits_{h\to 0^-} (h) \leq \lim\limits_{h\to 0^-} \left[-h\cos\left(\frac{1}{h}\right)\right] \leq \lim\limits_{h\to 0^-} (-h)$

$\Rightarrow 0 \leq \lim\limits_{h\to 0^-} \left[-h\cos\left(\frac{1}{h}\right)\right] \leq -0$

$\Rightarrow 0 \leq \lim\limits_{h\to 0^-} \left[-h\cos\left(\frac{1}{h}\right)\right] \leq 0$

Now, according to Squeeze/Sandwich theorem: $\mathbf{\lim\limits_{h\to 0^-} \left[-h\cos\left(\frac{1}{h}\right)\right] = 0}$

$\mathbf{f'_+(0) = \lim\limits_{h\to 0^-} \frac{0-f(0-h)}{h} = \lim\limits_{h\to 0^-} \frac{-f(-h)}{h} = 0}$

Now we've to find: Left Hand Derivative (L. H. D)

$$f'_-(0) = \mathbf{\lim\limits_{x\to 0^-} \frac{f(x)-0}{x-0} = \lim\limits_{x\to 0^-} \frac{0-f(x)}{0-x} = \lim\limits_{h\to 0^-} \frac{f(0+h)-0}{h} = \lim\limits_{h\to 0^+} \frac{0-f(0-h)}{h}}$$

Method 1:

$\mathbf{f'_-(0) = \lim\limits_{x\to 0^-} \frac{f(x)-0}{x-0} = \lim\limits_{x\to 0^-} \frac{0-f(x)}{0-x} = \lim\limits_{x\to 0^-} \frac{f(x)}{x}}$

Now, since, $f(x) = x^2\cos\left(\frac{1}{x}\right)$, when $x \neq 0$.

So, for $x \to 0^-$, $f(x) = x^2\cos\left(\frac{1}{x}\right)$

So, $\lim\limits_{x\to 0^-} \frac{f(x)}{x} = \lim\limits_{x\to 0^-} \frac{x^2\cos\left(\frac{1}{x}\right)}{x} = \lim\limits_{x\to 0^-} \left[x\cos\left(\frac{1}{x}\right)\right]$

Now, $\lim\limits_{x\to 0^-} x = 0$. But, $\lim\limits_{x\to 0^-} \left[\cos\left(\frac{1}{x}\right)\right]$ doesn't exist.

So, we can't use limit laws for product of two limits here.

Therefore, we'll use Squeeze/Sandwich theorem to determine $\lim\limits_{x\to 0^-} \left[x\cos\left(\frac{1}{x}\right)\right]$.

Now, we know: for $x \neq 0$:

$-1 \leq \cos\left(\frac{1}{x}\right) \leq 1$

$\Rightarrow -x \geq x\cos\left(\frac{1}{x}\right) \geq x$ [since, for, $x \to 0^-, x < 0$]

$\Rightarrow \lim\limits_{x\to 0^-} (-x) \geq \lim\limits_{x\to 0^-} \left[x\cos\left(\frac{1}{x}\right)\right] \geq \lim\limits_{x\to 0^-} (x)$

$\Rightarrow -0 \geq \lim\limits_{x\to 0^-} \left[x\cos\left(\frac{1}{x}\right)\right] \geq 0$

$\Rightarrow 0 \geq \lim\limits_{x\to 0^-} \left[x\cos\left(\frac{1}{x}\right)\right] \geq 0$

Now, according to Squeeze/Sandwich theorem: $\mathbf{\lim\limits_{x\to 0^-} \left[x\cos\left(\frac{1}{x}\right)\right] = 0}$

So, $\mathbf{f'_-(0) = \lim\limits_{x\to 0^-} \frac{f(x)-0}{x-0} = \lim\limits_{x\to 0^-} \frac{0-f(x)}{0-x} = \lim\limits_{x\to 0^-} \frac{f(x)}{x} = 0}$

Method 2:

$\mathbf{f'_-(0) = \lim\limits_{h\to 0^-} \frac{f(0+h)-0}{h} = \lim\limits_{h\to 0^-} \frac{f(h)}{h}}$

Now, since, $f(x) = x^2\cos\left(\frac{1}{x}\right)$, when $x \neq 0$.

So, for $h \to 0^-$, $f(h) = h^2\cos\left(\frac{1}{h}\right)$

So, $\lim\limits_{h\to 0^-} \frac{f(h)}{h} = \lim\limits_{h\to 0^-} \frac{h^2\cos\left(\frac{1}{h}\right)}{h} = \lim\limits_{h\to 0^-} \left[h\cos\left(\frac{1}{h}\right)\right]$

Now, $\lim\limits_{h\to 0^-} h = 0$. But, $\lim\limits_{h\to 0^-} \left[\cos\left(\frac{1}{h}\right)\right]$ doesn't exist.

So, we can't use limit laws for product of two limits here.

Therefore, we'll use Squeeze/Sandwich theorem to determine $\lim\limits_{h\to 0^-} \left[h\cos\left(\frac{1}{h}\right)\right]$.

Now, we know: for $h \neq 0$:

$-1 \leq \cos\left(\frac{1}{h}\right) \leq 1$

$\Rightarrow -h \geq h\cos\left(\frac{1}{h}\right) \geq h$ [since, for $h \to 0^-, h < 0$]

$\Rightarrow \lim\limits_{h\to 0^-} (-h) \geq \lim\limits_{h\to 0^-} \left[h\cos\left(\frac{1}{h}\right)\right] \geq \lim\limits_{h\to 0^-} (h)$

$\Rightarrow -0 \geq \lim\limits_{h\to 0^-} \left[h\cos\left(\frac{1}{h}\right)\right] \geq 0$

$\Rightarrow 0 \geq \lim\limits_{h\to 0^-} \left[h\cos\left(\frac{1}{h}\right)\right] \geq 0$

Now, according to Squeeze/Sandwich theorem: $\mathbf{\lim\limits_{h\to 0^-} \left[h\cos\left(\frac{1}{h}\right)\right] = 0}$

So, $\mathbf{f'_-(0) = \lim\limits_{h\to 0^-} \frac{f(0+h)-0}{h} = \lim\limits_{h\to 0^-} \frac{f(h)}{h} = 0}$

Method 3:

$\mathbf{f'_-(0) = \lim\limits_{h\to 0^+} \frac{0-f(0-h)}{h} = \lim\limits_{h\to 0^+} \frac{-f(-h)}{h}}$

Now, since, $f(x) = x^2\cos\left(\frac{1}{x}\right)$, when $x \neq 0$.

So, for $h \to 0^+ \Rightarrow -h \to 0^-$, $f(-h) = (-h)^2\cos\left(\frac{1}{-h}\right)$

So, $\lim\limits_{h\to0^+}\frac{-f(-h)}{h}$

$= \lim\limits_{h\to0^+}\frac{-(-h)^2\cos\left(\frac{1}{-h}\right)}{h}$

$= \lim\limits_{h\to0^+}\frac{-h^2\cos\left(\frac{1}{h}\right)}{h}$ [Since, $\cos(-\theta) = \cos\theta$]

$= \lim\limits_{h\to0^+}\left[-h\cos\left(\frac{1}{h}\right)\right]$

Now, $\lim\limits_{h\to0^+}(-h) = 0$. But, $\lim\limits_{h\to0^+}\left[\cos\left(\frac{1}{h}\right)\right]$ doesn't exist.

So, we can't use limit laws for product of two limits here.

Therefore, we'll use Squeeze/Sandwich theorem to determine $\lim\limits_{h\to0^+}\left[-h\cos\left(\frac{1}{h}\right)\right]$.

Now, we know: for $h \neq 0$:

$-1 \leq \cos\left(\frac{1}{h}\right) \leq 1$

$\Rightarrow -(-h) \geq -h\cos\left(\frac{1}{h}\right) \geq -h$ [since, for $h \to 0^+ \Rightarrow -h \to 0^-, -h < 0$]

$\Rightarrow \lim\limits_{h\to0^+}(h) \geq \lim\limits_{h\to0^+}\left[-h\cos\left(\frac{1}{h}\right)\right] \geq \lim\limits_{h\to0^+}(-h)$

$\Rightarrow 0 \geq \lim\limits_{h\to0^+}\left[-h\cos\left(\frac{1}{h}\right)\right] \geq -0$

$\Rightarrow 0 \geq \lim\limits_{h\to0^+}\left[-h\cos\left(\frac{1}{h}\right)\right] \geq 0$

Now, according to Squeeze/Sandwich theorem: $\boldsymbol{\lim\limits_{h\to0^+}\left[-h\cos\left(\frac{1}{h}\right)\right] = 0}$

$\boldsymbol{f'_-(0) = \lim\limits_{h\to0^+}\frac{0-f(0-h)}{h} = \lim\limits_{h\to0^+}\frac{-f(-h)}{h} = 0}$

Since, $f'_+(0) = f'_-(0) = 0$. Therefore, $f'(0)$ exists.

Now, $f'(0) = \lim\limits_{x\to0}\frac{f(x)-f(0)}{x-0} = \lim\limits_{x\to0}\frac{f(0)-f(x)}{0-x} = \lim\limits_{h\to0}\frac{f(0+h)-f(0)}{h} = \lim\limits_{h\to0}\frac{f(0)-f(0-h)}{h} = 0$

So, f(x) is differentiable at $x = 0$ and the derivative of f(x) at $x = 0$: $f'(0) = 0$

3. Find the differential coefficientof the following function with respect to x: [8]

$$(\sin x)^{\cos x} + (\cos x)^{\sin x}$$

Solution
See full solution here: https://www.emathhelp.net/calculators/calculus-1/derivative-calculator/?f=%28sin+%28x%29%29+%5E+%28cos+x%29+%2B+%28cos+%28x%29%29%5E+%28sin+%28x%29%29&order=1&var=x&p=

SOLUTION

The derivative of a sum/difference is the sum/difference of derivatives:

$$\left(\frac{d}{dx}\left(\sin^{\cos(x)}(x)+\cos^{\sin(x)}(x)\right)\right)=\left(\frac{d}{dx}\left(\sin^{\cos(x)}(x)\right)+\frac{d}{dx}\left(\cos^{\sin(x)}(x)\right)\right)$$

Use the formula $f^{g(x)}(x)=e^{g(x)\ln(f(x))}$ with $f(x)=\sin(x)$ and $g(x)=\cos(x)$ to rewrite the complex expression:

$$\left(\frac{d}{dx}\left(\sin^{\cos(x)}(x)\right)\right)+\frac{d}{dx}\left(\cos^{\sin(x)}(x)\right)=\left(\frac{d}{dx}\left(e^{\cos(x)\ln(\sin(x))}\right)\right)+\frac{d}{dx}\left(\cos^{\sin(x)}(x)\right)$$

The function $e^{\ln(\sin(x))\cos(x)}$ is the composition $f(g(x))$ of two functions $f(u)=e^{u}$ and $g(x)=\ln(\sin(x))\cos(x)$.

Apply the chain rule $\frac{d}{dx}(f(g(x)))=\frac{d}{du}(f(u))\frac{d}{dx}(g(x))$**:**

$$\left(\frac{d}{dx}\left(e^{\ln(\sin(x))\cos(x)}\right)\right)+\frac{d}{dx}\left(\cos^{\sin(x)}(x)\right)=\left(\frac{d}{du}(e^{u})\frac{d}{dx}(\ln(\sin(x))\cos(x))\right)+\frac{d}{dx}\left(\cos^{\sin(x)}(x)\right)$$

The derivative of the exponential is $\frac{d}{du}(e^{u})=e^{u}$**:**

$$\left(\frac{d}{du}(e^{u})\right)\frac{d}{dx}(\ln(\sin(x))\cos(x))+\frac{d}{dx}\left(\cos^{\sin(x)}(x)\right)=(e^{u})\frac{d}{dx}(\ln(\sin(x))\cos(x))+\frac{d}{dx}\left(\cos^{\sin(x)}(x)\right)$$

Return to the old variable:

$$e^{(u)}\frac{d}{dx}(\ln(\sin(x))\cos(x))+\frac{d}{dx}\left(\cos^{\sin(x)}(x)\right)=$$
$$e^{(\ln(\sin(x))\cos(x))}\frac{d}{dx}(\ln(\sin(x))\cos(x))+\frac{d}{dx}\left(\cos^{\sin(x)}(x)\right)=$$
$$\sin^{\cos(x)}(x)\frac{d}{dx}(\ln(\sin(x))\cos(x))+\frac{d}{dx}\left(\cos^{\sin(x)}(x)\right)$$

Apply the product rule $\frac{d}{dx}\left(f(x)g(x)\right) = \frac{d}{dx}\left(f(x)\right)g(x) + f(x)\frac{d}{dx}\left(g(x)\right)$ with $f(x) = \cos(x)$ and $g(x) = \ln(\sin(x))$:

$$\sin^{\cos(x)}(x)\left(\frac{d}{dx}\left(\ln(\sin(x))\cos(x)\right)\right) + \frac{d}{dx}\left(\cos^{\sin(x)}(x)\right) = \sin^{\cos(x)}(x)\left(\frac{d}{dx}\left(\cos(x)\right)\ln(\sin(x)) + \cos(x)\frac{d}{dx}\left(\ln(\sin(x))\right)\right) + \frac{d}{dx}\left(\cos^{\sin(x)}(x)\right)$$

The function $\ln(\sin(x))$ is the composition $f(g(x))$ of two functions $f(u) = \ln(u)$ and $g(x) = \sin(x)$.

Apply the chain rule $\frac{d}{dx}\left(f(g(x))\right) = \frac{d}{du}\left(f(u)\right)\frac{d}{dx}\left(g(x)\right)$:

$$\left(\ln(\sin(x))\frac{d}{dx}\left(\cos(x)\right) + \cos(x)\left(\frac{d}{dx}\left(\ln(\sin(x))\right)\right)\right)\sin^{\cos(x)}(x) + \frac{d}{dx}\left(\cos^{\sin(x)}(x)\right) = \left(\ln(\sin(x))\frac{d}{dx}\left(\cos(x)\right) + \cos(x)\left(\frac{d}{du}\left(\ln(u)\right)\frac{d}{dx}\left(\sin(x)\right)\right)\right)\sin^{\cos(x)}(x) + \frac{d}{dx}\left(\cos^{\sin(x)}(x)\right)$$

The derivative of the natural logarithm is $\frac{d}{du}\left(\ln(u)\right) = \frac{1}{u}$:

$$\left(\ln(\sin(x))\frac{d}{dx}\left(\cos(x)\right) + \cos(x)\left(\frac{d}{du}\left(\ln(u)\right)\right)\frac{d}{dx}\left(\sin(x)\right)\right)\sin^{\cos(x)}(x) + \frac{d}{dx}\left(\cos^{\sin(x)}(x)\right) = \left(\ln(\sin(x))\frac{d}{dx}\left(\cos(x)\right) + \cos(x)\left(\frac{1}{u}\right)\frac{d}{dx}\left(\sin(x)\right)\right)\sin^{\cos(x)}(x) + \frac{d}{dx}\left(\cos^{\sin(x)}(x)\right)$$

Return to the old variable:

$$\left(\ln\left(\sin\left(x\right)\right)\frac{d}{dx}\left(\cos\left(x\right)\right)+\frac{\cos\left(x\right)\frac{d}{dx}\left(\sin\left(x\right)\right)}{\left(u\right)}\right)\sin^{\cos\left(x\right)}\left(x\right)+\frac{d}{dx}\left(\cos^{\sin\left(x\right)}\left(x\right)\right)=\left(\ln\left(\sin\left(x\right)\right)\frac{d}{dx}\left(\cos\left(x\right)\right)+\frac{\cos\left(x\right)\frac{d}{dx}\left(\sin\left(x\right)\right)}{\left(\sin\left(x\right)\right)}\right)\sin^{\cos\left(x\right)}\left(x\right)+\frac{d}{dx}\left(\cos^{\sin\left(x\right)}\left(x\right)\right)$$

The derivative of the sine is $\frac{d}{dx}\left(\sin\left(x\right)\right)=\cos\left(x\right)$:

$$\left(\ln\left(\sin\left(x\right)\right)\frac{d}{dx}\left(\cos\left(x\right)\right)+\frac{\cos\left(x\right)\left(\frac{d}{dx}\left(\sin\left(x\right)\right)\right)}{\sin\left(x\right)}\right)\sin^{\cos\left(x\right)}\left(x\right)+\frac{d}{dx}\left(\cos^{\sin\left(x\right)}\left(x\right)\right)=\left(\ln\left(\sin\left(x\right)\right)\frac{d}{dx}\left(\cos\left(x\right)\right)+\frac{\cos\left(x\right)\left(\cos\left(x\right)\right)}{\sin\left(x\right)}\right)\sin^{\cos\left(x\right)}\left(x\right)+\frac{d}{dx}\left(\cos^{\sin\left(x\right)}\left(x\right)\right)$$

Use the formula $f^{g(x)}(x)=e^{g(x)\ln(f(x))}$ with $f(x)=\cos\left(x\right)$ and $g(x)=\sin\left(x\right)$ to rewrite the complex expression:

$$\left(\ln\left(\sin\left(x\right)\right)\frac{d}{dx}\left(\cos\left(x\right)\right)+\frac{\cos^{2}\left(x\right)}{\sin\left(x\right)}\right)\sin^{\cos\left(x\right)}\left(x\right)+\left(\frac{d}{dx}\left(\cos^{\sin\left(x\right)}\left(x\right)\right)\right)=\left(\ln\left(\sin\left(x\right)\right)\frac{d}{dx}\left(\cos\left(x\right)\right)+\frac{\cos^{2}\left(x\right)}{\sin\left(x\right)}\right)\sin^{\cos\left(x\right)}\left(x\right)+\left(\frac{d}{dx}\left(e^{\sin\left(x\right)\ln\left(\cos\left(x\right)\right)}\right)\right)$$

The function $e^{\ln\left(\cos\left(x\right)\right)\sin\left(x\right)}$ is the composition $f(g(x))$ of two functions $f(u)=e^{u}$ and $g(x)=\ln\left(\cos\left(x\right)\right)\sin\left(x\right)$.

Apply the chain rule $\frac{d}{dx}\left(f(g(x))\right)=\frac{d}{du}\left(f(u)\right)\frac{d}{dx}\left(g(x)\right)$:

$$\left(\ln\left(\sin\left(x\right)\right)\frac{d}{dx}\left(\cos\left(x\right)\right)+\frac{\cos^{2}\left(x\right)}{\sin\left(x\right)}\right)\sin^{\cos\left(x\right)}\left(x\right)+\left(\frac{d}{dx}\left(e^{\ln\left(\cos\left(x\right)\right)\sin\left(x\right)}\right)\right)=\left(\ln\left(\sin\left(x\right)\right)\frac{d}{dx}\left(\cos\left(x\right)\right)+\frac{\cos^{2}\left(x\right)}{\sin\left(x\right)}\right)\sin^{\cos\left(x\right)}\left(x\right)+\left(\frac{d}{du}\left(e^{u}\right)\frac{d}{dx}\left(\ln\left(\cos\left(x\right)\right)\sin\left(x\right)\right)\right)$$

The derivative of the exponential is $\frac{d}{du}\left(e^{u}\right)=e^{u}$:

$$\left(\ln\left(\sin\left(x\right)\right)\frac{d}{dx}\left(\cos\left(x\right)\right)+\frac{\cos^{2}\left(x\right)}{\sin\left(x\right)}\right)\sin^{\cos\left(x\right)}\left(x\right)+\left(\frac{d}{du}\left(e^{u}\right)\right)\frac{d}{dx}\left(\ln\left(\cos\left(x\right)\right)\sin\left(x\right)\right)=\left(\ln\left(\sin\left(x\right)\right)\frac{d}{dx}\left(\cos\left(x\right)\right)+\frac{\cos^{2}\left(x\right)}{\sin\left(x\right)}\right)\sin^{\cos\left(x\right)}\left(x\right)+\left(e^{u}\right)\frac{d}{dx}\left(\ln\left(\cos\left(x\right)\right)\sin\left(x\right)\right)$$

Return to the old variable:

$$\left(\ln\left(\sin\left(x\right)\right)\frac{d}{dx}\left(\cos\left(x\right)\right)+\frac{\cos^{2}\left(x\right)}{\sin\left(x\right)}\right)\sin^{\cos\left(x\right)}\left(x\right)+e^{\left(u\right)}\frac{d}{dx}\left(\ln\left(\cos\left(x\right)\right)\sin\left(x\right)\right)=\left(\ln\left(\sin\left(x\right)\right)\frac{d}{dx}\left(\cos\left(x\right)\right)+\frac{\cos^{2}\left(x\right)}{\sin\left(x\right)}\right)\sin^{\cos\left(x\right)}\left(x\right)+e^{\left(\ln\left(\cos\left(x\right)\right)\sin\left(x\right)\right)}\frac{d}{dx}\left(\ln\left(\cos\left(x\right)\right)\sin\left(x\right)\right)=\left(\ln\left(\sin\left(x\right)\right)\frac{d}{dx}\left(\cos\left(x\right)\right)+\frac{\cos^{2}\left(x\right)}{\sin\left(x\right)}\right)\sin^{\cos\left(x\right)}\left(x\right)+\cos^{\sin\left(x\right)}\left(x\right)\frac{d}{dx}\left(\ln\left(\cos\left(x\right)\right)\sin\left(x\right)\right)$$

Apply the product rule $\frac{d}{dx}\left(f(x)g(x)\right)=\frac{d}{dx}\left(f(x)\right)g(x)+f(x)\frac{d}{dx}\left(g(x)\right)$ **with** $f(x)=\ln\left(\cos\left(x\right)\right)$ **and** $g(x)=\sin\left(x\right)$:

$$\left(\ln\left(\sin\left(x\right)\right)\frac{d}{dx}\left(\cos\left(x\right)\right)+\frac{\cos^{2}\left(x\right)}{\sin\left(x\right)}\right)\sin^{\cos\left(x\right)}\left(x\right)+\cos^{\sin\left(x\right)}\left(x\right)\left(\frac{d}{dx}\left(\ln\left(\cos\left(x\right)\right)\sin\left(x\right)\right)\right)=\left(\ln\left(\sin\left(x\right)\right)\frac{d}{dx}\left(\cos\left(x\right)\right)+\frac{\cos^{2}\left(x\right)}{\sin\left(x\right)}\right)\sin^{\cos\left(x\right)}\left(x\right)+\cos^{\sin\left(x\right)}\left(x\right)\left(\frac{d}{dx}\left(\ln\left(\cos\left(x\right)\right)\right)\sin\left(x\right)+\ln\left(\cos\left(x\right)\right)\frac{d}{dx}\left(\sin\left(x\right)\right)\right)$$

The derivative of the cosine is $\frac{d}{dx}\left(\cos\left(x\right)\right)=-\sin\left(x\right)$:

$$\left(\ln\left(\sin\left(x\right)\right)\left(\frac{d}{dx}\left(\cos\left(x\right)\right)\right)+\frac{\cos^{2}\left(x\right)}{\sin\left(x\right)}\right)\sin^{\cos\left(x\right)}\left(x\right)+\left(\ln\left(\cos\left(x\right)\right)\frac{d}{dx}\left(\sin\left(x\right)\right)+\sin\left(x\right)\frac{d}{dx}\left(\ln\left(\cos\left(x\right)\right)\right)\right)\cos^{\sin\left(x\right)}\left(x\right)=\left(\ln\left(\sin\left(x\right)\right)\left(-\sin\left(x\right)\right)+\frac{\cos^{2}\left(x\right)}{\sin\left(x\right)}\right)\sin^{\cos\left(x\right)}\left(x\right)+\left(\ln\left(\cos\left(x\right)\right)\frac{d}{dx}\left(\sin\left(x\right)\right)+\sin\left(x\right)\frac{d}{dx}\left(\ln\left(\cos\left(x\right)\right)\right)\right)\cos^{\sin\left(x\right)}\left(x\right)$$

The function $\ln(\cos(x))$ is the composition $f(g(x))$ of two functions $f(u) = \ln(u)$ and $g(x) = \cos(x)$.

Apply the chain rule $\frac{d}{dx}(f(g(x))) = \frac{d}{du}(f(u))\frac{d}{dx}(g(x))$:

$$\left(-\ln(\sin(x))\sin(x) + \frac{\cos^2(x)}{\sin(x)}\right)\sin^{\cos(x)}(x) + \left(\ln(\cos(x))\frac{d}{dx}(\sin(x)) + \sin(x)\left(\frac{d}{dx}(\ln(\cos(x)))\right)\right)\cos^{\sin(x)}(x) = \left(-\ln(\sin(x))\sin(x) + \frac{\cos^2(x)}{\sin(x)}\right)\sin^{\cos(x)}(x) + \left(\ln(\cos(x))\frac{d}{dx}(\sin(x)) + \sin(x)\left(\frac{d}{du}(\ln(u))\frac{d}{dx}(\cos(x))\right)\right)\cos^{\sin(x)}(x)$$

The derivative of the natural logarithm is $\frac{d}{du}(\ln(u)) = \frac{1}{u}$:

$$\left(-\ln(\sin(x))\sin(x) + \frac{\cos^2(x)}{\sin(x)}\right)\sin^{\cos(x)}(x) + \left(\ln(\cos(x))\frac{d}{dx}(\sin(x)) + \sin(x)\left(\frac{d}{du}(\ln(u))\right)\frac{d}{dx}(\cos(x))\right)\cos^{\sin(x)}(x) = \left(-\ln(\sin(x))\sin(x) + \frac{\cos^2(x)}{\sin(x)}\right)\sin^{\cos(x)}(x) + \left(\ln(\cos(x))\frac{d}{dx}(\sin(x)) + \sin(x)\left(\frac{1}{u}\right)\frac{d}{dx}(\cos(x))\right)\cos^{\sin(x)}(x)$$

Return to the old variable:

$$\left(-\ln(\sin(x))\sin(x) + \frac{\cos^2(x)}{\sin(x)}\right)\sin^{\cos(x)}(x) + \left(\ln(\cos(x))\frac{d}{dx}(\sin(x)) + \frac{\sin(x)\frac{d}{dx}(\cos(x))}{(u)}\right)\cos^{\sin(x)}(x) = \left(-\ln(\sin(x))\sin(x) + \frac{\cos^2(x)}{\sin(x)}\right)\sin^{\cos(x)}(x) + \left(\ln(\cos(x))\frac{d}{dx}(\sin(x)) + \frac{\sin(x)\frac{d}{dx}(\cos(x))}{(\cos(x))}\right)\cos^{\sin(x)}(x)$$

The derivative of the cosine is $\frac{d}{dx}(\cos(x)) = -\sin(x)$:

$$\left(-\ln(\sin(x))\sin(x) + \frac{\cos^2(x)}{\sin(x)}\right)\sin^{\cos(x)}(x) + \left(\ln(\cos(x))\frac{d}{dx}(\sin(x)) + \frac{\sin(x)\left(\frac{d}{dx}(\cos(x))\right)}{\cos(x)}\right)\cos^{\sin(x)}(x) = \left(-\ln(\sin(x))\sin(x) + \frac{\cos^2(x)}{\sin(x)}\right)\sin^{\cos(x)}(x) + \left(\ln(\cos(x))\frac{d}{dx}(\sin(x)) + \frac{\sin(x)(-\sin(x))}{\cos(x)}\right)\cos^{\sin(x)}(x)$$

The derivative of the sine is $\frac{d}{dx}\left(\sin(x)\right) = \cos(x)$:

$$\left(-\ln(\sin(x))\sin(x) + \frac{\cos^2(x)}{\sin(x)}\right)\sin^{\cos(x)}(x) + \left(\ln(\cos(x))\left(\frac{d}{dx}(\sin(x))\right) - \frac{\sin^2(x)}{\cos(x)}\right)\cos^{\sin(x)}(x) = \left(-\ln(\sin(x))\sin(x) + \frac{\cos^2(x)}{\sin(x)}\right)\sin^{\cos(x)}(x) + \left(\ln(\cos(x))(\cos(x)) - \frac{\sin^2(x)}{\cos(x)}\right)\cos^{\sin(x)}(x)$$

Simplify:

$$\left(-\ln(\sin(x))\sin(x) + \frac{\cos^2(x)}{\sin(x)}\right)\sin^{\cos(x)}(x) + \left(\ln(\cos(x))\cos(x) - \frac{\sin^2(x)}{\cos(x)}\right)\cos^{\sin(x)}(x) = \frac{-\left(\ln(\sin(x))\sin^2(x) - \cos^2(x)\right)\sin^{\cos(x)}(x)\cos(x) + \left(\ln(\cos(x))\cos^2(x) - \sin^2(x)\right)\sin(x)\cos^{\sin(x)}(x)}{\sin(x)\cos(x)}$$

Thus, $\frac{d}{dx}\left(\sin^{\cos(x)}(x) + \cos^{\sin(x)}(x)\right) = \frac{-\left(\ln(\sin(x))\sin^2(x) - \cos^2(x)\right)\sin^{\cos(x)}(x)\cos(x) + \left(\ln(\cos(x))\cos^2(x) - \sin^2(x)\right)\sin(x)\cos^{\sin(x)}(x)}{\sin(x)\cos(x)}$.

ANSWER

$$\frac{d}{dx}\left(\sin^{\cos(x)}(x) + \cos^{\sin(x)}(x)\right) = \frac{-\left(\ln(\sin(x))\sin^2(x) - \cos^2(x)\right)\sin^{\cos(x)}(x)\cos(x) + \left(\ln(\cos(x))\cos^2(x) - \sin^2(x)\right)\sin(x)\cos^{\sin(x)}(x)}{\sin(x)\cos(x)}$$ A

4. Find all critical and stationary points of the function $f(x) = 3x^{5/3} - 15x^{2/3}$. [8]
Also find the relative extrema.

Solution
See full solution using the following online solvers: **Find maximum and minimum value of y=x^3+6x^2-15x+7 calculator** **Extreme Points Calculator - Find Extreme and Saddle Points** **Derivative calculator** **Derivative Calculator - eMathHelp** **Function Calculator - eMathHelp** **Evaluate Calculator - eMathHelp**
Critical and stationary points
To find critical points, first compute **f′(x)** **https://www.emathhelp.net/calculators/calculus-1/derivative-calculator/?f=3x%5E%285%2F3%29+-+15x%5E%282%2F3%29&order=1&var=x&p=**

Find $\frac{d}{dx}\left(3x^{\frac{5}{3}}-15x^{\frac{2}{3}}\right)$.

SOLUTION

The derivative of a sum/difference is the sum/difference of derivatives:

$$\left(\frac{d}{dx}\left(3x^{\frac{5}{3}}-15x^{\frac{2}{3}}\right)\right)=\left(\frac{d}{dx}\left(3x^{\frac{5}{3}}\right)-\frac{d}{dx}\left(15x^{\frac{2}{3}}\right)\right)$$

Apply the constant multiple rule $\frac{d}{dx}\left(cf(x)\right)=c\frac{d}{dx}\left(f(x)\right)$ **with** $c=15$ **and** $f(x)=x^{\frac{2}{3}}$:

$$-\left(\frac{d}{dx}\left(15x^{\frac{2}{3}}\right)\right)+\frac{d}{dx}\left(3x^{\frac{5}{3}}\right)=-\left(15\frac{d}{dx}\left(x^{\frac{2}{3}}\right)\right)+\frac{d}{dx}\left(3x^{\frac{5}{3}}\right)$$

Apply the constant multiple rule $\frac{d}{dx}\left(cf(x)\right)=c\frac{d}{dx}\left(f(x)\right)$ **with** $c=3$ **and** $f(x)=x^{\frac{5}{3}}$:

$$\left(\frac{d}{dx}\left(3x^{\frac{5}{3}}\right)\right)-15\frac{d}{dx}\left(x^{\frac{2}{3}}\right)=\left(3\frac{d}{dx}\left(x^{\frac{5}{3}}\right)\right)-15\frac{d}{dx}\left(x^{\frac{2}{3}}\right)$$

Apply the power rule $\frac{d}{dx}\left(x^n\right)=nx^{n-1}$ **with** $n=\frac{5}{3}$:

$$3\left(\frac{d}{dx}\left(x^{\frac{5}{3}}\right)\right)-15\frac{d}{dx}\left(x^{\frac{2}{3}}\right)=3\left(\frac{5x^{\frac{2}{3}}}{3}\right)-15\frac{d}{dx}\left(x^{\frac{2}{3}}\right)$$

Apply the power rule $\frac{d}{dx}\left(x^n\right)=nx^{n-1}$ **with** $n=\frac{2}{3}$:

$$5x^{\frac{2}{3}}-15\left(\frac{d}{dx}\left(x^{\frac{2}{3}}\right)\right)=5x^{\frac{2}{3}}-15\left(\frac{2}{3\sqrt[3]{x}}\right)$$

Simplify:

$$5x^{\frac{2}{3}}-\frac{10}{\sqrt[3]{x}}=\frac{5\left(x-2\right)}{\sqrt[3]{x}}$$

Thus, $\frac{d}{dx}\left(3x^{\frac{5}{3}}-15x^{\frac{2}{3}}\right)=\frac{5(x-2)}{\sqrt[3]{x}}$.

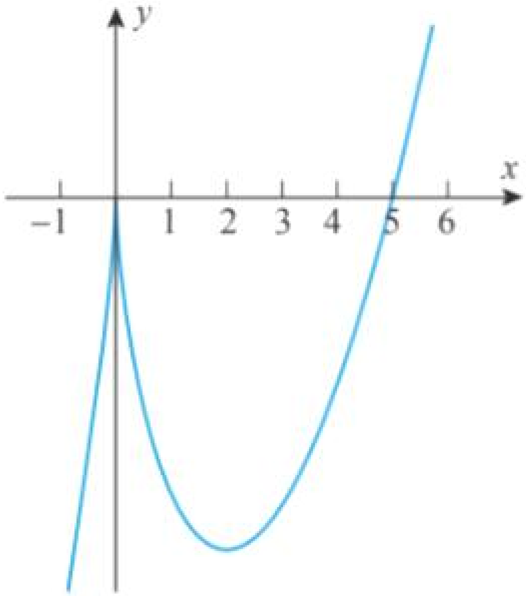
ANSWER

$\frac{d}{dx}\left(3x^{\frac{5}{3}}-15x^{\frac{2}{3}}\right)=\frac{5(x-2)}{\sqrt[3]{x}}$ A

The function f is continuous everywhere and its derivative is

$$f'(x)=5x^{2/3}-10x^{-1/3}=5x^{-1/3}(x-2)=\frac{5(x-2)}{x^{1/3}}$$

We see from this that $f'(x) = 0$ if $x = 2$ and $f'(x)$ is undefined if $x = 0$. Thus $x = 0$ and $x = 2$ are critical points and $x = 2$ is a stationary point. This is consistent with the graph of f shown in Figure 4.2.5. ◄



$y = 3x^{5/3} - 15x^{2/3}$

▲ **Figure 4.2.5**

Relative extrema

Method 1:

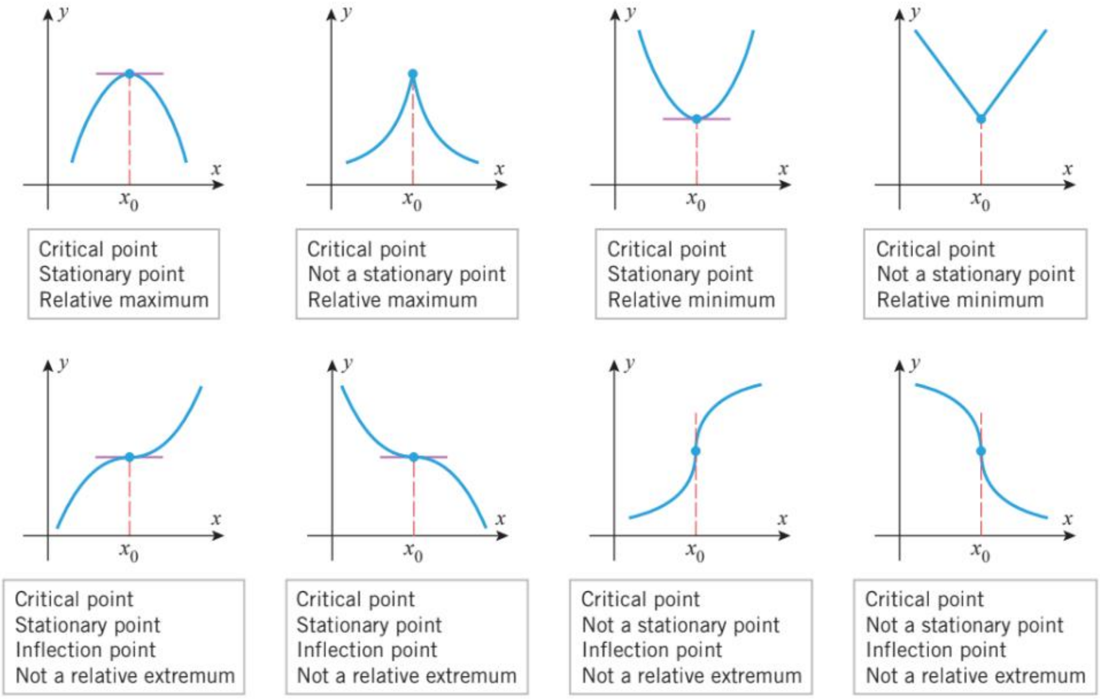
4.2.3 THEOREM (*First Derivative Test*) *Suppose that f is continuous at a critical point x_0.*

(*a*) *If $f'(x) > 0$ on an open interval extending left from x_0 and $f'(x) < 0$ on an open interval extending right from x_0, then f has a relative maximum at x_0.*

(*b*) *If $f'(x) < 0$ on an open interval extending left from x_0 and $f'(x) > 0$ on an open interval extending right from x_0, then f has a relative minimum at x_0.*

(*c*) *If $f'(x)$ has the same sign on an open interval extending left from x_0 as it does on an open interval extending right from x_0, then f does not have a relative extremum at x_0.*

A function f has a relative extremum at those critical points where f' changes sign.



▲ **Figure 4.2.6**

► **Example 4** We showed in Example 3 that the function $f(x) = 3x^{5/3} - 15x^{2/3}$ has critical points at $x = 0$ and $x = 2$. Figure 4.2.5 suggests that f has a relative maximum at $x = 0$ and a relative minimum at $x = 2$. Confirm this using the first derivative test.

Table 4.2.1

INTERVAL	$5(x-2)/x^{1/3}$	$f'(x)$
$x < 0$	$(-)/(-)$	$+$
$0 < x < 2$	$(-)/(+)$	$-$
$x > 2$	$(+)/(+)$	$+$

Solution. We showed in Example 3 that

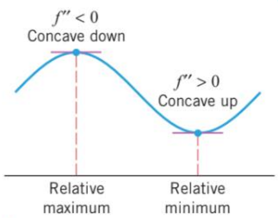
$$f'(x) = \frac{5(x-2)}{x^{1/3}}$$

A sign analysis of this derivative is shown in Table 4.2.1. The sign of f' changes from $+$ to $-$ at $x = 0$, so there is a relative maximum at that point. The sign changes from $-$ to $+$ at $x = 2$, so there is a relative minimum at that point. ◄

Method 2:

SECOND DERIVATIVE TEST

There is another test for relative extrema that is based on the following geometric observation: A function f has a relative maximum at a stationary point if the graph of f is concave down on an open interval containing that point, and it has a relative minimum if it is concave up (Figure 4.2.7).



▲ **Figure 4.2.7**

4.2.4 THEOREM (*Second Derivative Test*) *Suppose that f is twice differentiable at the point x_0.*

(a) If $f'(x_0) = 0$ and $f''(x_0) > 0$, then f has a relative minimum at x_0.

(b) If $f'(x_0) = 0$ and $f''(x_0) < 0$, then f has a relative maximum at x_0.

(c) If $f'(x_0) = 0$ and $f''(x_0) = 0$, then the test is inconclusive; that is, f may have a relative maximum, a relative minimum, or neither at x_0.

Now, we'll find f ''(x) :
https://www.emathhelp.net/calculators/calculus-1/derivative-calculator/?f=3x%5E%285%2F3%29+-+15x%5E%282%2F3%29&order=2&var=&p=

Find $\frac{d^2}{dx^2}\left(3x^{\frac{5}{3}} - 15x^{\frac{2}{3}}\right)$.

SOLUTION

Find the first derivative $\frac{d}{dx}\left(3x^{\frac{5}{3}} - 15x^{\frac{2}{3}}\right)$

The derivative of a sum/difference is the sum/difference of derivatives:

$$\left(\frac{d}{dx}\left(3x^{\frac{5}{3}} - 15x^{\frac{2}{3}}\right)\right) = \left(\frac{d}{dx}\left(3x^{\frac{5}{3}}\right) - \frac{d}{dx}\left(15x^{\frac{2}{3}}\right)\right)$$

Apply the constant multiple rule $\frac{d}{dx}\left(cf(x)\right) = c\frac{d}{dx}\left(f(x)\right)$ **with** $c = 15$ **and** $f(x) = x^{\frac{2}{3}}$:

$$-\left(\frac{d}{dx}\left(15x^{\frac{2}{3}}\right)\right) + \frac{d}{dx}\left(3x^{\frac{5}{3}}\right) = -\left(15\frac{d}{dx}\left(x^{\frac{2}{3}}\right)\right) + \frac{d}{dx}\left(3x^{\frac{5}{3}}\right)$$

Apply the constant multiple rule $\frac{d}{dx}\left(cf(x)\right) = c\frac{d}{dx}\left(f(x)\right)$ **with** $c = 3$ **and** $f(x) = x^{\frac{5}{3}}$:

$$\left(\frac{d}{dx}\left(3x^{\frac{5}{3}}\right)\right) - 15\frac{d}{dx}\left(x^{\frac{2}{3}}\right) = \left(3\frac{d}{dx}\left(x^{\frac{5}{3}}\right)\right) - 15\frac{d}{dx}\left(x^{\frac{2}{3}}\right)$$

Apply the power rule $\frac{d}{dx}\left(x^n\right) = nx^{n-1}$ **with** $n = \frac{5}{3}$:

$$3\left(\frac{d}{dx}\left(x^{\frac{5}{3}}\right)\right) - 15\frac{d}{dx}\left(x^{\frac{2}{3}}\right) = 3\left(\frac{5x^{\frac{2}{3}}}{3}\right) - 15\frac{d}{dx}\left(x^{\frac{2}{3}}\right)$$

Apply the power rule $\frac{d}{dx}\left(x^n\right) = nx^{n-1}$ **with** $n = \frac{2}{3}$:

$$5x^{\frac{2}{3}} - 15\left(\frac{d}{dx}\left(x^{\frac{2}{3}}\right)\right) = 5x^{\frac{2}{3}} - 15\left(\frac{2}{3\sqrt[3]{x}}\right)$$

Simplify:

$$5x^{\frac{2}{3}} - \frac{10}{\sqrt[3]{x}} = \frac{5\left(x-2\right)}{\sqrt[3]{x}}$$

Thus, $\frac{d}{dx}\left(3x^{\frac{5}{3}} - 15x^{\frac{2}{3}}\right) = \frac{5(x-2)}{\sqrt[3]{x}}$.

Next, $\frac{d^2}{dx^2}\left(3x^{\frac{5}{3}} - 15x^{\frac{2}{3}}\right) = \frac{d}{dx}\left(\frac{5(x-2)}{\sqrt[3]{x}}\right)$

Apply the constant multiple rule $\frac{d}{dx}\left(cf(x)\right) = c\frac{d}{dx}\left(f(x)\right)$ with $c = 5$ and $f(x) = \frac{x-2}{\sqrt[3]{x}}$:

$$\left(\frac{d}{dx}\left(\frac{5\left(x-2\right)}{\sqrt[3]{x}}\right)\right) = \left(5\frac{d}{dx}\left(\frac{x-2}{\sqrt[3]{x}}\right)\right)$$

Apply the quotient rule $\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{\frac{d}{dx}\left(f(x)\right)g(x) - f(x)\frac{d}{dx}\left(g(x)\right)}{g^2(x)}$ with $f(x) = x - 2$ and $g(x) = \sqrt[3]{x}$:

$$5\left(\frac{d}{dx}\left(\frac{x-2}{\sqrt[3]{x}}\right)\right) = 5\left(\frac{\frac{d}{dx}\left(x-2\right)\sqrt[3]{x} - \left(x-2\right)\frac{d}{dx}\left(\sqrt[3]{x}\right)}{\left(\sqrt[3]{x}\right)^2}\right)$$

The derivative of a sum/difference is the sum/difference of derivatives:

$$\frac{5\left(\sqrt[3]{x}\left(\frac{d}{dx}\left(x-2\right)\right) - \left(x-2\right)\frac{d}{dx}\left(\sqrt[3]{x}\right)\right)}{x^{\frac{2}{3}}} = \frac{5\left(\sqrt[3]{x}\left(\frac{d}{dx}\left(x\right) - \frac{d}{dx}\left(2\right)\right) - \left(x-2\right)\frac{d}{dx}\left(\sqrt[3]{x}\right)\right)}{x^{\frac{2}{3}}}$$

The derivative of a constant is 0:

$$\frac{5\left(\sqrt[3]{x}\left(-\left(\frac{d}{dx}\left(2\right)\right) + \frac{d}{dx}\left(x\right)\right) - \left(x-2\right)\frac{d}{dx}\left(\sqrt[3]{x}\right)\right)}{x^{\frac{2}{3}}} = \frac{5\left(\sqrt[3]{x}\left(-\left(0\right) + \frac{d}{dx}\left(x\right)\right) - \left(x-2\right)\frac{d}{dx}\left(\sqrt[3]{x}\right)\right)}{x^{\frac{2}{3}}}$$

Apply the power rule $\frac{d}{dx}\left(x^n\right) = nx^{n-1}$ with $n = 1$, in other words, $\frac{d}{dx}\left(x\right) = 1$:

$$\frac{5\left(\sqrt[3]{x}\left(\frac{d}{dx}\left(x\right)\right) - \left(x-2\right)\frac{d}{dx}\left(\sqrt[3]{x}\right)\right)}{x^{\frac{2}{3}}} = \frac{5\left(\sqrt[3]{x}\left(1\right) - \left(x-2\right)\frac{d}{dx}\left(\sqrt[3]{x}\right)\right)}{x^{\frac{2}{3}}}$$

Apply the power rule $\frac{d}{dx}\left(x^n\right) = nx^{n-1}$ with $n = \frac{1}{3}$:

$$\frac{5\left(\sqrt[3]{x} - \left(x-2\right)\left(\frac{d}{dx}\left(\sqrt[3]{x}\right)\right)\right)}{x^{\frac{2}{3}}} = \frac{5\left(\sqrt[3]{x} - \left(x-2\right)\left(\frac{1}{3x^{\frac{2}{3}}}\right)\right)}{x^{\frac{2}{3}}}$$

Simplify:

$$\frac{5\left(\sqrt[3]{x}-\frac{x-2}{3x^{\frac{2}{3}}}\right)}{x^{\frac{2}{3}}}=\frac{10\left(x+1\right)}{3x^{\frac{4}{3}}}$$

Thus, $\frac{d}{dx}\left(\frac{5(x-2)}{\sqrt[3]{x}}\right)=\frac{10(x+1)}{3x^{\frac{4}{3}}}$.

Therefore, $\frac{d^2}{dx^2}\left(3x^{\frac{5}{3}}-15x^{\frac{2}{3}}\right)=\frac{10(x+1)}{3x^{\frac{4}{3}}}$.

ANSWER

$$\frac{d^2}{dx^2}\left(3x^{\frac{5}{3}}-15x^{\frac{2}{3}}\right)=\frac{10(x+1)}{3x^{\frac{4}{3}}}$$

Evaluating f ''(x) at critical point x=2 means f ''(2):

https://www.emathhelp.net/calculators/algebra-2/evaluate-calculator/?i=10*%28x+%2B+1%29%2F%283*x%5E%284%2F3%29%29&v=x%3D2

Your input: evaluate $\frac{10\left(x+1\right)}{3x^{\frac{4}{3}}}$ **when** $x=2$.

ANSWER

Answer: $\frac{5\cdot 2^{\frac{2}{3}}}{2}\approx 3.9685026299205$.

Since, f ''(2)>0. So, f(x) has a relative minimum at x=2.

Evaluating f ''(x) at critical point x=0 means f ''(0):
Since, f '(0) and f ''(0) does not exist. So, second derivative test can’t be applied here. We’ve to use **Method 1** at critical point x=0. Checking sign of f '(0) before and after x=0 we can see that f(x) has a relative maximum at

5. Verify the hypothesis of the Mean Value Theorem for the function [8]

$$f(x) = \sqrt{4x - 3}$$

over the interval [1,3]. Also find the value of 'c' in that interval that satisfy the conclusion of the theorem.

Solution
See full solution here: **https://www.emathhelp.net/en/calculators/calculus-1/mean-value-theorem-calculator/?f=sqrt%284x-3%29&a=1&b=3**

The Mean Value Theorem states that for a continuous and differentiable function $f(x)$ on the interval $[a, b]$ there exists such number c from the interval (a, b), that $f'(c) = \frac{f(b) - f(a)}{b - a}$.

First, evaluate the function at the endpoints of the interval:

$f(3) = 3$

$f(1) = 1$

Next, find the derivative: $f'(c) = \frac{2}{\sqrt{4c - 3}}$ (for steps, see derivative calculator).

Form the equation: $\frac{2}{\sqrt{4c - 3}} = \frac{(3) - (1)}{(3) - (1)}$

Simplify: $\frac{2}{\sqrt{4c - 3}} = 1$

Solve the equation on the given interval: $c = \frac{7}{4}$

Answer: $\frac{7}{4} = 1.75$

6. Find the Maclaurin polynomials P_0, P_1, P_2, P_3 of the function $f(x) = e^x \cos x$. [8]

Solution
See full solution here: https://www.emathhelp.net/calculators/calculus-1/taylor-and-maclaurin-series-calculator/?f=%28e%5Ex+%29*%28cos+x%29&p=0&n=3&v=

Using the fact that $P_n = P_{n-1} + n - th\ order\ terms$, We can get P_0, P_1, P_2, P_3 sequentially.

Your input: calculate the Taylor (Maclaurin) series of $e^x \cos(x)$ up to $n = 3$

A Maclaurin series is given by $f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!} x^k$

In our case, $f(x) \approx P(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!} x^k = \sum_{k=0}^{3} \frac{f^{(k)}(a)}{k!} x^k$

So, what we need to do to get the desired polynomial is to calculate the derivatives, evaluate them at the given point, and plug the results into the given formula.

$f^{(0)}(x) = f(x) = e^x \cos(x)$

Evaluate the function at the point: $f(0) = 1$

1. Find the 1st derivative: $f^{(1)}(x) = \left(f^{(0)}(x)\right)' = (e^x \cos(x))' = \sqrt{2} e^x \cos\left(x + \frac{\pi}{4}\right)$ (steps can be seen here).

   Evaluate the 1st derivative at the given point: $(f(0))' = 1$

2. Find the 2nd derivative: $f^{(2)}(x) = \left(f^{(1)}(x)\right)' = \left(\sqrt{2} e^x \cos\left(x + \frac{\pi}{4}\right)\right)' = -2e^x \sin(x)$ (steps can be seen here).

   Evaluate the 2nd derivative at the given point: $(f(0))'' = 0$

3. Find the 3rd derivative: $f^{(3)}(x) = \left(f^{(2)}(x)\right)' = (-2e^x \sin(x))' = -2\sqrt{2} e^x \sin\left(x + \frac{\pi}{4}\right)$ (steps can be seen here).

   Evaluate the 3rd derivative at the given point: $(f(0))''' = -2$

Now, use the calculated values to get a polynomial:

$$f(x) \approx \frac{1}{0!}x^0 + \frac{1}{1!}x^1 + \frac{0}{2!}x^2 + \frac{-2}{3!}x^3$$

Finally, after simplifying we get the final answer:

$$f(x) \approx P(x) = 1 + x - \frac{1}{3}x^3$$

Answer: the Taylor (Maclaurin) series of $e^x \cos(x)$ up to $n = 3$ is $e^x \cos(x) \approx P(x) = 1 + x - \frac{1}{3}x^3$

7. Let, $f(x) = \frac{x-1}{\sqrt{x^2}}$ and $\lim\limits_{x\to+\infty} f(x) = a$ and $\lim\limits_{x\to-\infty} f(x) = b$. [8]

The straight lines $y = a$ and $y = b$ are called **HORIZONTAL ASYMPTOTES** of the graph of $y = f(x) = \frac{x-1}{\sqrt{x^2}}$.

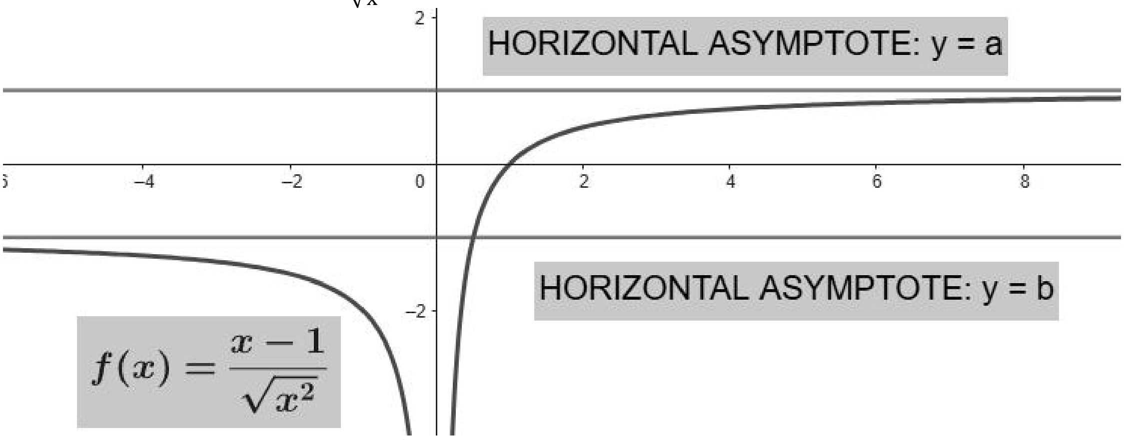
Find the value of $(a - b)$.

Hints: $\sqrt{x^2} = |x| = \begin{cases} -x, & x < 0 \\ x, & x \geq 0 \end{cases}$

Solution

In figure, $f(x) = \frac{x-1}{\sqrt{x^2}}$ and $\lim\limits_{x\to+\infty} f(x) = a$ and $\lim\limits_{x\to-\infty} f(x) = b$.

The straight lines $y = a$ and $y = b$ are called **HORIZONTAL ASYMPTOTES** of the graph of $y = f(x) = \frac{x-1}{\sqrt{x^2}}$.



Hints: $\sqrt{x^2} = |x| = \begin{cases} -x, & x < 0 \\ x, & x \geq 0 \end{cases}$

Since, $x \to +\infty \Rightarrow x > 0$.

So, $a = \lim\limits_{x\to+\infty} f(x) = \lim\limits_{x\to+\infty}\left(\frac{x-1}{\sqrt{x^2}}\right) = \lim\limits_{x\to+\infty}\left(\frac{x-1}{|x|}\right) = \lim\limits_{x\to+\infty}\left(\frac{x-1}{x}\right)$

And, since, $x \to -\infty \Rightarrow x < 0$.

So, $b = \lim\limits_{x\to-\infty} f(x) = \lim\limits_{x\to-\infty}\left(\frac{x-1}{\sqrt{x^2}}\right) = \lim\limits_{x\to-\infty}\left(\frac{x-1}{|x|}\right) = \lim\limits_{x\to-\infty}\left(\frac{x-1}{-x}\right)$

Now, a = $\lim\limits_{x\to+\infty} f(x) = \lim\limits_{x\to+\infty}\left(\frac{x-1}{\sqrt{x^2}}\right) = \lim\limits_{x\to+\infty}\left(\frac{x-1}{|x|}\right) = \lim\limits_{x\to+\infty}\left(\frac{x-1}{x}\right)$

See Solution here:
https://www.emathhelp.net/calculators/calculus-1/limit-calculator/?f=%28x-1%29%2F+sqrt+%28x%5E2%29&var=&val=inf&dir=

Your input: find $\lim\limits_{x\to\infty}\frac{x-1}{x}$

Rewrite the function:

$$\lim_{x\to\infty}\frac{x-1}{x} = \lim_{x\to\infty}\left(1-\frac{1}{x}\right)$$

The limit of a sum/difference is the sum/difference of limits:

$$\lim_{x\to\infty}\left(1-\frac{1}{x}\right) = \left(\lim_{x\to\infty} 1 - \lim_{x\to\infty}\frac{1}{x}\right)$$

The limit of a constant is equal to the constant:

$$-\lim_{x\to\infty}\frac{1}{x} + \lim_{x\to\infty} 1 = -\lim_{x\to\infty}\frac{1}{x} + 1$$

The limit of a quotient is the quotient of limits:

$$1-\lim_{x\to\infty}\frac{1}{x} = 1-\frac{\lim_{x\to\infty} 1}{\lim_{x\to\infty} x}$$

The limit of a constant is equal to the constant:

$$1-\frac{\lim_{x\to\infty} 1}{\lim_{x\to\infty} x} = 1-\frac{1}{\lim_{x\to\infty} x}$$

Constant divided by a very big number equals 0:

$$1-1\frac{1}{\lim_{x\to\infty} x} = 1-(0)$$

Therefore,

$$\lim_{x\to\infty}\frac{x-1}{x} = 1$$

Threfore, a = $\lim\limits_{x\to+\infty} f(x) = \lim\limits_{x\to+\infty}\left(\frac{x-1}{\sqrt{x^2}}\right) = \lim\limits_{x\to+\infty}\left(\frac{x-1}{|x|}\right) = \lim\limits_{x\to+\infty}\left(\frac{x-1}{x}\right) = 1$

And, b = $\lim\limits_{x\to-\infty} f(x) = \lim\limits_{x\to-\infty}\left(\frac{x-1}{\sqrt{x^2}}\right) = \lim\limits_{x\to-\infty}\left(\frac{x-1}{|x|}\right) = \lim\limits_{x\to-\infty}\left(\frac{x-1}{-x}\right)$

See Solution here:
https://www.emathhelp.net/calculators/calculus-1/limit-calculator/?f=%28x-1%29%2F+sqrt+%28x%5E2%29&var=&val=-inf&dir=

Your input: find $\lim\limits_{x\to-\infty}\left(-\frac{x-1}{x}\right)$

Apply the constant multiple rule $\lim\limits_{x\to-\infty} cf(x) = c\lim\limits_{x\to-\infty} f(x)$ **with** $c=-1$ **and** $f(x)=\frac{x-1}{x}$:

$$\lim_{x\to-\infty}\left(-\frac{x-1}{x}\right) = \left(-\lim_{x\to-\infty}\frac{x-1}{x}\right)$$

Rewrite the function:

$$-\lim_{x\to-\infty}\frac{x-1}{x} = -\lim_{x\to-\infty}\left(1-\frac{1}{x}\right)$$

The limit of a sum/difference is the sum/difference of limits:

$$-\lim_{x\to-\infty}\left(1-\frac{1}{x}\right) = -\left(\lim_{x\to-\infty}1 - \lim_{x\to-\infty}\frac{1}{x}\right)$$

The limit of a constant is equal to the constant:

$$\lim_{x\to-\infty}\frac{1}{x} - \lim_{x\to-\infty}1 = \lim_{x\to-\infty}\frac{1}{x} - 1$$

The limit of a quotient is the quotient of limits:

$$-1+\lim_{x\to-\infty}\frac{1}{x} = -1+\frac{\lim_{x\to-\infty}1}{\lim_{x\to-\infty}x}$$

The limit of a constant is equal to the constant:

$$-1+\frac{\lim_{x\to-\infty}1}{\lim_{x\to-\infty}x} = -1+\frac{1}{\lim_{x\to-\infty}x}$$

Constant divided by a very big number equals 0**:**

$$-1+1\frac{1}{\lim_{x\to-\infty}x} = -1+(0)$$

$$\lim_{x\to-\infty}\left(-\frac{x-1}{x}\right) = -1$$

And, therefore, b = $\lim\limits_{x\to-\infty} f(x) = \lim\limits_{x\to-\infty}\left(\frac{x-1}{\sqrt{x^2}}\right) = \lim\limits_{x\to-\infty}\left(\frac{x-1}{|x|}\right) = \lim\limits_{x\to-\infty}\left(\frac{x-1}{-x}\right) =-1$

So, the value of $\mathbf{a-b = 1-(-1) = 1+1 = 2}$